

FINAL EXAMINATION SYLLABUS

MAM2000W Module 2AC (Advanced Calculus)

Wednesday 14 June 2023, 17:00 - 19:00

University of Cape Town
Department of Mathematics and Applied Mathematics

The exam is very similar in style to the class tests, except that it is now 2 hours and hence has more questions, and the syllabus is a combination of what we have covered in lectures and tutorials this semester.

As in the class tests, no proofs of theorems nor derivations of formulae will be tested.

You may use a non-programmable calculator during the exam.

The exam will consist of questions requiring full answers as well as multiple choice questions. There is no negative marking on the multiple choice questions.

Course notes:

1. **Vector functions-Curves-I and -II:** Vector functions and their domains, limits, continuity, differentiability, and integrals. Parametric curves and showing a curve is regular. Tangent line to a regular curve, the unit tangent vector, and the arclength of a curve. Curvature, normal and binormal vectors, and the osculating, normal, and rectifying planes.
2. **Domain-Limit-Continuity:** Functions of several variables: domain, level sets and graphs, limits at a point, and continuity.
3. **Differentiability-I and -II:** Functions of several variables: partial derivatives, differentiability at a point, the tangent plane to a graph; the role of the gradient w.r.t. increase and decrease, and the directional derivative; the chain rule; higher-order partial derivatives.
4. **Parametric surfaces-I and -II:** Showing a surface is regular, tangent plane to a parametric surface, tangent plane to a level surface. Polar, cylindrical and spherical coordinates.
5. **Maximum-Minimum-I, -II, and -IV:** Unconstrained optimization (critical points, nature of critical points using $D(a,b)$); constrained optimization with the method of the Lagrange multiplier for one equality constraint, and over a closed, bounded domain.
6. **Multiple-Integrals-I, -II, -III, and -IV:** The double integral in a rectangular domain, evaluation of double integrals over non-rectangular domains. Change of variables in double integrals, evaluation of a double integral through the use of polar coordinates. Evaluation of triple integrals over rectangular and non-rectangular regions, change of variables in triple integrals via cylindrical and spherical coordinates.
7. **Vector-Calculus-I, -II, -III, and -IV:** Conservative vector fields, line integrals over scalar-valued functions, line integrals over vector fields, curve orientation, Green's theorem. Curl and divergence of a vector field. Surface integrals (including the area of a surface) of both parametric surfaces and surfaces described by graphs.

Stokes' and Divergence Theorems will NOT be examined!

Questions and their solutions:

- Tutorials 1-13
- Exercise sheets
- Class tests 1 and 2